NET731 Week 2

Network Design Report

Diploma in IT - Network Design and Administration

Module: NET731

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1. Introduction

This report presents the Week 2 practical work for the NET731 Azure capstone project. The task focused on building the network foundation of the Azure environment by creating a Virtual Network, configuring an address space with multiple subnets, and creating a Network Security Group for subnet-level protection.

2. Objective of Week 2

The objective of Week 2 was to design and configure an Azure network that can support later project tasks such as virtual machine deployment, identity management, storage configuration, and security monitoring. The completed network structure is organised, scalable, and suitable for the next stages of the project.

3. Virtual Network Configuration

A Virtual Network named CTU-NET731-VNet was created inside the resource group CTU-20231149-RG. The network was deployed in the South Africa North region using the Azure for Students subscription. This Virtual Network provides the main communication boundary for Azure resources used in the project.

4. Address Space and Subnets

The Virtual Network was configured with the address space 10.0.0.0/16. This range was selected because it provides a large number of available IP addresses and allows logical separation of services into dedicated subnets.

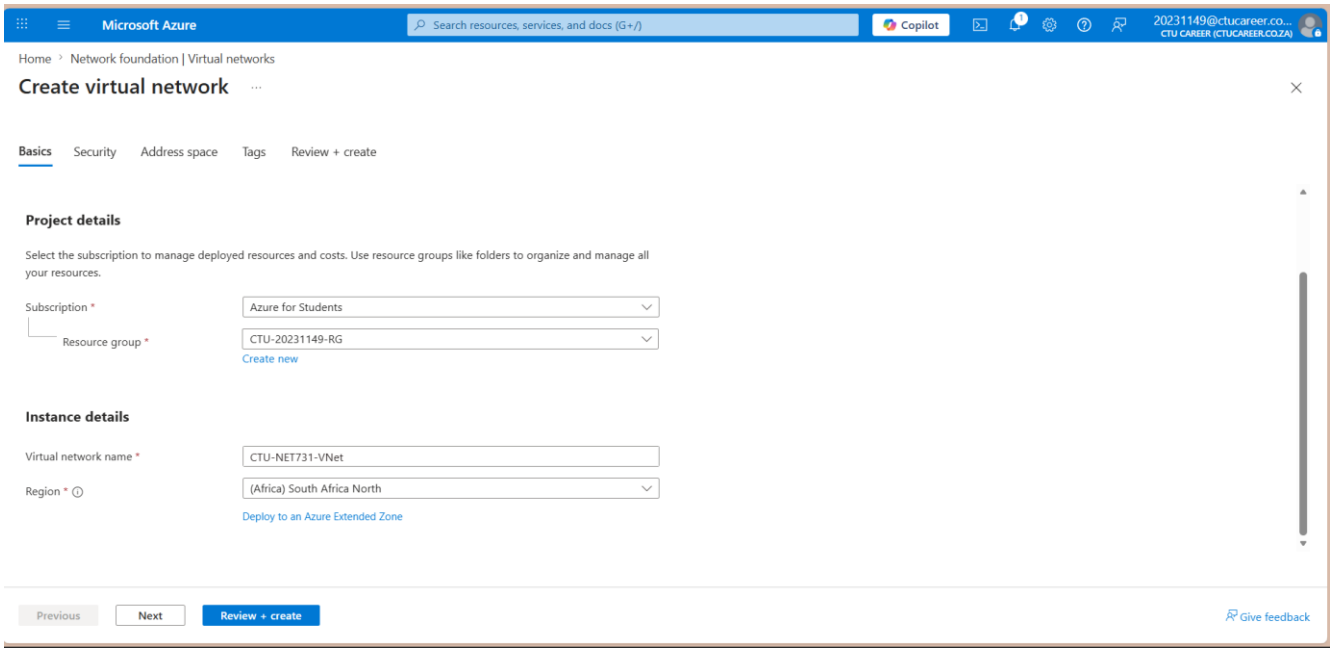
Subnet Name	Address Range	Purpose
FrontendSubnet	10.0.1.0/24	Frontend or user-facing resources
BackendSubnet	10.0.2.0/24	Internal services and backend resources
ManagementSubnet	10.0.3.0/24	Administrative and management-related resources

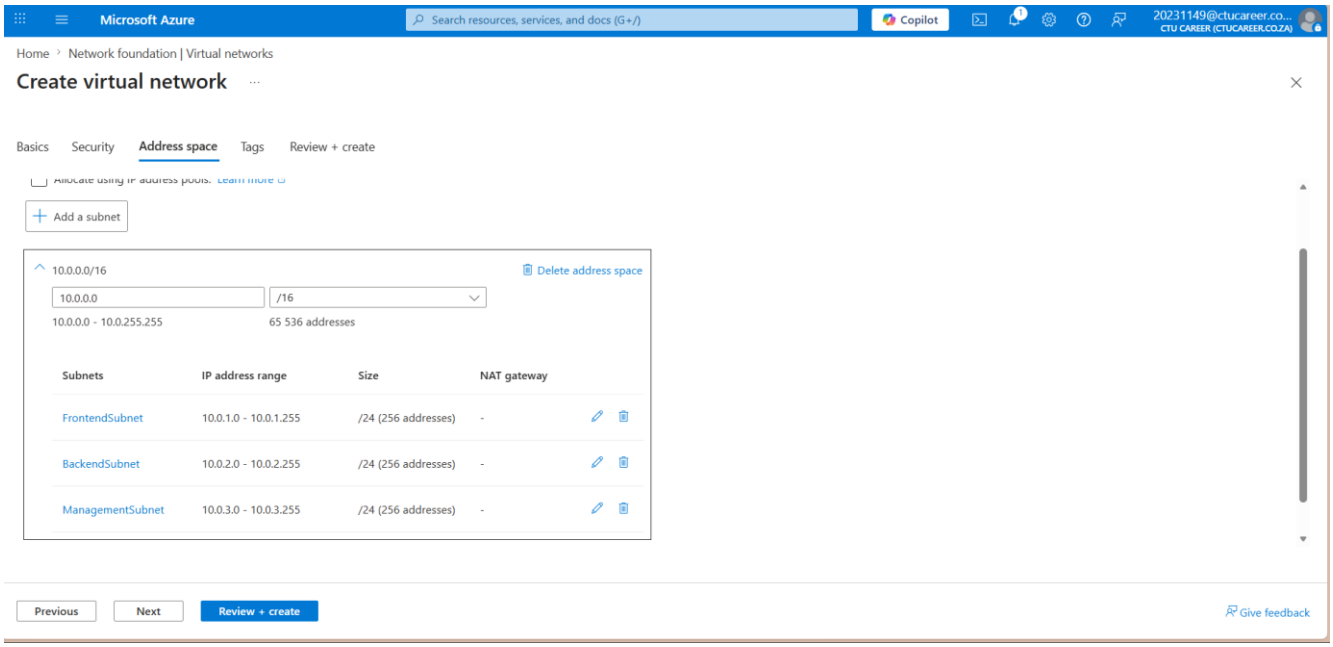
Creating separate subnets improves organisation and makes it easier to manage network traffic and apply security controls more effectively.

5. Network Security Group Configuration

A Network Security Group named CTU-NET731-NSG was created in the same resource group and region. The NSG contains Azure default inbound and outbound rules and was associated with FrontendSubnet in the CTU-NET731-VNet Virtual Network. This association applies traffic filtering at subnet level and helps protect Azure resources from unnecessary or unauthorised traffic.

6. The screenshots below provide evidence of the completed Week 2 practical steps.





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Home > CTU-NET731-VNet-1776771060191 | Overview

CTU-NET731-VNet
Virtual network

What is the best way to secure my virtual networks? | Suggest ways to lower costs in managing virtual networks. | Check health of virtual network

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Move | Delete | Refresh | Give feedback

Essentials | JSON View

Resource group (move)	: CTU-20231149-RG	Address space	: 10.0.0.0/16
Location (move)	: South Africa North	Subnets	: 3 subnets
Subscription (move)	: Azure for Students	DNS servers	: Azure provided DNS service
Subscription ID	: 6c105eae-dd4e-485e-aa12-16556ce8e975	BGP community string	: Configure
		Virtual network ID	: 937d4d0d-8ee4-4fa6-97d9-05361298b70b

Tags (edit) : Add tags

Topology | Properties | **Capabilities** | Recommendations | Tutorials

DDoS protection
Configure additional protection from distributed denial of service attacks.
Not configured

Azure Firewall
Protect your network with a stateful L3-L7 firewall.
Not configured

Peerings
Seamlessly connect two or more virtual networks.
Not configured

Microsoft Defender for Cloud
Strengthen the security posture of your environment.

Private endpoints

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Home > CreateNetworkSecurityGroupBladeV2-20260421133815 | Overview

CTU-NET731-NSG
Network security group

Retrieve detailed information for troubleshooting security rules | How do I create an alert to track firewall metric failures | Diagnose connectivity issues related to this security group

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
- Monitoring
- Automation
- Help

Move | Delete | Refresh | Give feedback

Essentials | JSON View

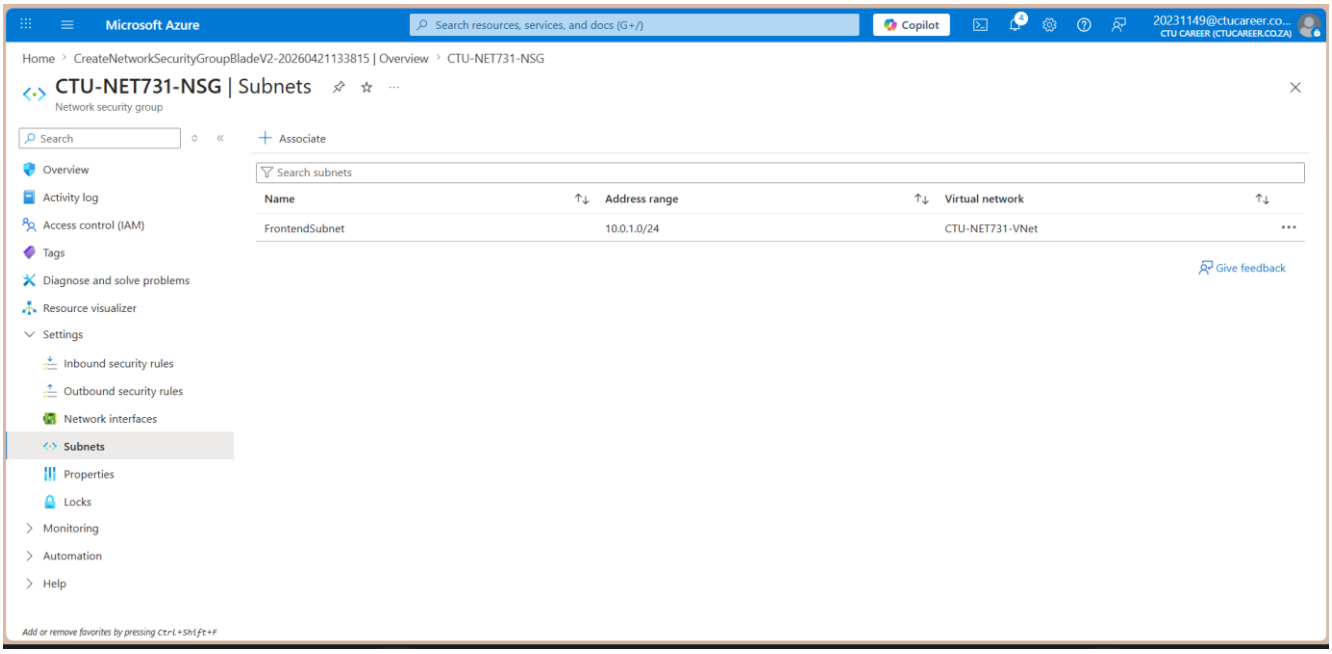
Resource group (move)	: CTU-20231149-RG	Custom security rules	: 0 inbound, 0 outbound
Location	: South Africa North	Associated with	: 0 subnets, 0 network interfaces
Subscription (move)	: Azure for Students		
Subscription ID	: 6c105eae-dd4e-485e-aa12-16556ce8e975		

Tags (edit) : Add tags

Filter by name | Port == all | Protocol == all | Source == all | Destination == all | Action == all

Priority	Name	Port	Protocol	Source	Destination	Action
Inbound Security Rules						
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerI---	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound Security Rules						
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

Add or remove favorites by pressing Ctrl+Shift+F



7. Conclusion

The Azure network foundation now includes a properly configured Virtual Network, a clearly structured address space, three working subnets, and a Network Security Group attached to FrontendSubnet. This provides a secure and organised environment for the next weekly activities in the Azure capstone project.

8. References

Microsoft. (n.d.) Virtual Network documentation. Available at: https://learn.microsoft.com/azure/virtual-network/

Microsoft. (n.d.) Create a virtual network using the Azure portal. Available at: https://learn.microsoft.com/azure/virtual-network/quickstart-create-virtual-network

Microsoft. (n.d.) Network Security Groups documentation. Available at: https://learn.microsoft.com/azure/virtual-network/network-security-groups-overview

Microsoft. (n.d.) Manage network security groups. Available at: https://learn.microsoft.com/azure/virtual-network/manage-network-security-group

GitHub Docs. (n.d.) Create a repository. Available at: https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository